

Education

University Of Maryland, College Park

College Park, MD

Computer Science, Undergraduate Student

Aug. 2022 - May. 2026

- **Relevant Coursework:** Introduction to Algorithms, Design and Analysis of Computer Algorithms, Introduction to Machine Learning, Introduction to Artificial Intelligence, Introduction to Computer Systems, Advanced Data Structures, Applied Probability and Statistics, Discrete Structures, Linear Algebra, Operating Systems

Experience

Yellow.ai

Remote / Charlotte, NC

AI Engineering Intern

Aug 2025 - Sept 2025

- Designed and prototyped an AI-powered content moderation tool capable of classifying photos and videos as “appropriate” or “inappropriate,” with customizable thresholds defined by users.
- Leveraged machine learning libraries such as scikit-learn to build and train models for multimedia classification, integrating them into a scalable prototype for CDN filtering.
- Collaborated with product teams to align technical implementation with customer experience goals, ensuring flexibility and adaptability of the moderation system.

Huntington Ingalls Industries - Mission Technologies Division

Hanover, MD

Summer Intern

June 2025 - Aug. 2025

- Iterated upon the Atop Log Parser and Visualizer tool to benchmark and improve Minotaur map rendering technology, focusing on thread management, memory leak detection, and system stability for mission-critical operations.
- Implemented performance monitoring and debugging strategies to ensure reliable rendering under high-load conditions.
- Explored machine learning approaches to predict potential system failures when processing large-scale inputs, enhancing proactive fault detection capabilities.

Huntington Ingalls Industries - Mission Technologies Division

Hanover, MD

Summer Intern

June 2024 - Aug. 2024

- Contributed to the development of Minotaur, a specialized platform integrating sensors, cameras, radar, and communication equipment into a unified system for Air and Marine Operations.
- Enhanced software functionality of Minotaur by implementing validation checks, such as ensuring user inputs do not exceed critical thresholds, significantly improving system stability and user experience.
- Expanded Minotaur’s capabilities by designing and integrating a new function to filter signals based on specific properties, addressing critical gaps in signal processing functionalities.
- Applied expertise in parsing and analyzing system performance data using atop to optimize software performance and troubleshoot operational issues effectively.

Qure.AI

Charlotte, NC

Summer Intern

May 2021 - Aug. 2021

- Studied convolutional neural networks, AI image recognition, and open-source machine learning architectures to build an AI-based image recognition program to detect medical abnormalities, such as the impact of COVID-19 on the lungs, captured in x-ray and MRI images, as well as size, location and growth rates of brain tumors.
- Gathered, arranged, and corrected research data of over 30,000 x-ray and MRI images, to create representative graphs and charts highlighting results for presentations.
- Implemented advanced Machine Learning concepts such as back-propagation and rectified linear activation functions, which both increase the accuracy and efficiency of the AI model, thereby achieving a 98% detection accuracy rate for the final model.
- Implemented an interactive UI for medical professionals to input multiple images at a time and quickly get results.

Projects

PureMark

Ongoing Development

- Cross-platform Markdown editor with live PDF preview, enabling professional document editing locally without requiring a system LaTeX installation.
- Engineered a Rust backend (Tauri 2) that bundles Pandoc and Tectonic, allowing seamless PDF/DOCX/LaTeX export on Linux, Windows, and macOS.
- Optimized performance through debounced re-renders and lazy rendering.
- Solved platform-specific challenges like WebKitGTK security taints and Pandoc binary path resolution by implementing dynamic shimming and cross-platform build logic.

Ryujinx Configurator

Ongoing Development

- Created a controller-navigable utility for Steam Deck/Linux, enabling in-gamepad binding directly from Steam Big Picture Mode without leaving gaming context.
- Reverse-engineered a non-standard controller GUID format used by the Ryujinx emulator, implementing byte-order conversion to ensure compatibility across hardware and streaming platforms.
- Designed a state-machine UI driven by SDL input, with atomic file writes and automatic backup logic to safely manage emulator configuration without corruption risk.
- Built a diagnostic overlay for debugging controller mismatches, improving setup time for couch co-op sessions from minutes to seconds.

PhishNet AI

April 2025

- Chrome web extension protecting users from phishing and malicious websites, integrating VirusTotal API and Gemini AI for threat explanation.
- Implemented automated domain blocking to prevent access to identified threats, reducing security risk during browsing sessions.
- Overcame Chrome Manifest V3 restrictions by designing secure API routing workarounds for real-time threat detection without breaking browser security.

Atop Log Parser and Visualizer

June 2024

- Developed a C++ program to parse mission log files for CPU and RAM performance data, supporting system monitoring in mission-critical environments.
- Created interactive charts and graphs using d3.js and HTML/JavaScript, enabling real-time visualization of system health data.
- Optimized parsing algorithms to handle large log files efficiently, reducing analysis time and improving monitoring accuracy for high-load systems.

MediCheck AI

April 2024

- Developed an AI-driven pre-diagnosis tool at UMD's BitCamp 2024, enabling users to input symptoms, medical history, and lifestyle factors via an intuitive web interface.
- Implemented RAG architecture to retrieve relevant medical knowledge from external knowledge bases, grounding LLM responses in verified medical literature and improving diagnostic accuracy.
- Powered by MedLLaMa2 and a Retrieval-Augmented Generation pipeline to provide preliminary medical diagnoses and health insights based on user inputs.
- Collaborated with a team to design a scalable architecture, integrating React frontend and Python-based APIs for backend inference.

Technical Skills

Languages: Java, Python, C, C++, JavaScript, TypeScript, HTML, CSS, Rust

Developer Tools: GitHub, Git, Google Cloud, Firebase, ClearCase, Jira, PyCharm, VS Code, Docker

Technologies/Frameworks: PyTorch, Pandas, SQL, Numpy, Jupyter Notebook, UNIX/Linux, React JS, Tauri, SDL, Pandoc, LaTeX

Concepts: REST APIs, CI/CD, Microservices, Linux System Administration, WebAssembly